Joonho Lee

Director, Neuromeka AI Lab

On military duty (until 2027.4.30., Technical Research Personnel)

Robotics researcher and hands-on systems engineer, combining robot learning research with end-to-end development and real-world deployment. PhD from ETH Zurich's Robotic Systems Lab, with a focus on legged robotics; now leading AI research for industrial robots at Neuromeka.

joonho.lee@neuromeka.com · Google Scholar

Experience

2024 – now	**Neuromeka Co., Ltd.** Director, AI Group · Military service
2022 – 2024	**RIVR** Conceived and developed the initial RL-based control and navigation stack

Education

2018 – 2024	**ETH Zurich, Robotic Systems Lab** PhD, Mechanical Engineering
2017 – 2018	**ETH Zurich** MSc Robotics, Systems and Control · Rank 1/76
2012 – 2016	**POSTECH** BSc Mechanical Engineering · Rank 2/17

Funding

2024 – 2029	**Development of AI-based medical and care humanoid robot** International Joint R&D Program, KIAT, Ministry of Trade, Industry and Energy Planned the project and lead its execution Partners: Georgia Tech (Sehoon Ha et al.), ETRI, KAIST ≈ US$7M (KRW 10B) over 5 years

Selected publications

2026	**Proprioceptive reflex learning for contact-rich manipulation** Quan Nguyen, Yunho Kim, Joonho Lee Submitted to ICRA 2027
2026	**Learning-augmented robotic automation for real-world manufacturing** Yunho Kim, Quan Nguyen, Taewhan Kim, Youngjin Heo, Joonho Lee arXiv:2604.22235 · under review at Nature Communications
2025	**Learning fast, tool-aware collision avoidance for collaborative robots** Joonho Lee, Yunho Kim, Seokjoon Kim, Quan Nguyen, Youngjin Heo IEEE RA-L
2025	**Learning-based legged locomotion: State of the art and future perspectives** Sehoon Ha, Joonho Lee, Michiel van de Panne, Zhaoming Xie, Wenhao Yu, Majid Khadiv IJRR 44(8)
2024	**Learning to walk in confined spaces using 3D representation** Takahiro Miki, Joonho Lee, Lorenz Wellhausen, Marco Hutter ICRA · Best Paper Award in Service Robotics

2024 **Scalable multi-robot cooperation for multi-goal tasks using reinforcement learning**
Joonho Lee*, Tianxu An*, Marko Bjelonic, Flavio De Vincenti, Marco Hutter
IEEE RA-L

2024 **Learning robust autonomous navigation and locomotion for wheeled-legged robots**
Joonho Lee, Marko Bjelonic, Alexander Reske, Lorenz Wellhausen, Takahiro Miki, Marco Hutter
Science Robotics 9(89)

2024 **When monkeys meet an ANYmal robot in the wild**
Charlotte Canteloup, Joonho Lee, Samuel Zimmermann, Morgane Alvino, Markus Montenegro, Marco Hutter, Erica van de Waal
bioRxiv 2024.08.13.607714

2022 **Learning robust perceptive locomotion for quadrupedal robots in the wild**
Takahiro Miki, Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, Marco Hutter
Science Robotics 7(62)

2020 **Learning quadrupedal locomotion over challenging terrain**
Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, Marco Hutter
Science Robotics 5(47) · IEEE RAS TC-MBOR Best Paper Award

Student & mentored papers

2026 **Proprioceptive reflex learning for contact-rich manipulation**
Quan Nguyen, Yunho Kim, Joonho Lee
Submitted to ICRA 2027

2026 **Learning-augmented robotic automation for real-world manufacturing**
Yunho Kim, Quan Nguyen, Taewhan Kim, Youngjin Heo, Joonho Lee
arXiv:2604.22235 · under review at Nature Communications

2025 **Gait optimization for underwater legged robots using data-driven hydrodynamic modeling and reinforcement learning**
Seokyong Song, Taesik Kim, Seungmin Kim, Joonho Lee, Son-Cheol Yu
IEEE RA-L

2024 **Scalable multi-robot cooperation for multi-goal tasks using reinforcement learning**
Joonho Lee*, Tianxu An*, Marko Bjelonic, Flavio De Vincenti, Marco Hutter
IEEE RA-L

2024 **Exploring constrained reinforcement learning algorithms for quadrupedal locomotion**
Joonho Lee*, Lukas Schroth*, Victor Klemm, Marko Bjelonic, Alexander Reske, Marco Hutter
IROS

2023 **Learning-based design and control for quadrupedal robots with parallel-elastic actuators**
Filip Bjelonic, Joonho Lee, Philip Arm, Dhionis Sako, Davide Tateo, Jan Peters, Marco Hutter
IEEE RA-L

2023 **Advanced skills through multiple adversarial motion priors in reinforcement learning**
Eric Vollenweider, Marko Bjelonic, Victor Klemm, Nikita Rudin, Joonho Lee, Marco Hutter
ICRA

2022 **Meta reinforcement learning for optimal design of legged robots**
Álvaro Belmonte-Baeza, Joonho Lee, Giorgio Valsecchi, Marco Hutter
IEEE RA-L

2020 **DeepGait: Planning and control of quadrupedal gaits using deep reinforcement learning**
Vassilios Tsounis, Mitja Alge, Joonho Lee, Farbod Farshidian, Marco Hutter
IEEE RA-L

Awards

2025	**ETH Zurich Medal for outstanding doctoral thesis** ETH Zurich
2024	**Best Paper Award in Service Robotics** IEEE ICRA
2020	**Best Paper Award, TC on Model-Based Optimization for Robotics** IEEE Robotics & Automation Society
2019	**ETH Zurich Medal for outstanding master's thesis** ETH Zurich
2019	**Willi Studer Prize (Best Graduate), Robotics, Systems and Control** D-MAVT, ETH Zurich

Invited talks & lectures

22 TALKS

2026.09.22	**From Paper to Factory Floor: Building Smart Robots for Industry** — ME Seminar, KAIST (KR)
2025.11.28	**Teaching Robots to Walk and Work** — CSE/GSAI Seminar, POSTECH (KR)
2025.06.04	**Training Robots to Move and Work** — IRIM Seminar, Georgia Institute of Technology (US)
2025.02.10	**Elevating Robotic Performance through Learned Autonomy** — ME Seminar, SNU (KR)
2024.10.15	**Advancing the Autonomy of Legged Robots** — KAIST AI (KR)
2024.01.02	**Learning locomotion and autonomous navigation for legged robots** — SUSTech (CN)
2023.12.29	**Learning locomotion and autonomous navigation for legged robots** — Zhejiang University (CN)
2023.08.02	**Navigating real-world with legged robots** — Georgia Institute of Technology (US)
2022.11.10	**Robust quadrupedal locomotion via RL** — Krafton (KR)
2022.11.08	**Toward the real-world applications of autonomous walking robots** — POSTECH (KR)
2022.11.03	**Developing legged robots for outdoor use** — Agency for Defense Development (ADD) (KR)
2022.04.25	**RL for quadrupedal locomotion** — DLR (DE)
2022.01.12	**Learning quadrupedal locomotion over challenging terrain** — RILAB, Korea University (KR)
2021.07.12	**Learning quadrupedal locomotion over challenging terrain** — CRL, ETH Zurich (CH)
2021.07.02	**Learning quadrupedal locomotion over challenging terrain** — Telluride Neuromorphic Cognition Engineering Workshop (US)
2021.05.21	**Learning quadrupedal locomotion over challenging terrain** [video] — NCCR Robotics Seminar (CH)
2021.02.26	**Learning quadrupedal locomotion over challenging terrain** [video] — PAIR, University of Toronto (CA)
2021.02.12	**인공지능기술을 활용한 4족보행로봇 제어** — AI Winter School, Korean Society of Mechanical Engineers (KSME) (KR)
2020.11.10	**Learning quadrupedal locomotion over challenging terrain** — CSIRO (AU)
2020.08.05	**Learning legged locomotion – walking through a forest** — Intel Labs (US)
2019.02.13	**시뮬레이션에서의 심층강화학습을 이용한 보행로봇 제어** — Korea Institute of Industrial Technology (KR)
2019.02.11	**시뮬레이션에서의 심층강화학습을 이용한 보행로봇 제어** — Agency for Defense Development (ADD) (KR)